Xuechun Yu

Urban Researcher & Planner

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Research Interests

Neighborhood Environments & Child Development; Spatial Inequality; Children's Outdoor Spaces; Environmental Justice; Spatial Analysis; Urban Soundscapes & Multimodal GeoAI

Education

The University of Hong Kong — Hong Kong, China
Master of Urban Design — *Aug 2021 – Dec 2022*
Thesis: *Analyzing Urban Spatial Structure of Retail Land Use Relative to Residential Land Use: A Case Study of Hong Kong* (Grade: A−)
Thesis Advisor: Prof. Alain Chiaradia

Dalian University of Technology — Dalian, China
B.E. in Urban and Rural Planning (5-year program) — *Sep 2013 – Jun 2018*

Publications and Work in Progress

Manuscripts

Xuechun Yu, Bowen Liao, Xiaoyu Li, and Yuhao Kang. *Mapping Urban Soundscape Perception from Street-View Imagery: An Image-to-Sound Framework Applied to Detroit.* Manuscript in preparation.

Xiaohao Yang, **Xuechun Yu**, Hanbing Liang, and Yuhao Kang. *From Desired Soundscapes to Streets: A Cross-Modal Mapping Framework Searching for a Desired Acoustic Experience.* Under review at *Journal of Urban Design*.

Conference Abstracts

Xuechun Yu. *"For a Younger Crowd?": Quantifying the Domain-Specific Divergence of Aging Resources in Gentrifying New York Neighborhoods.* Abstract accepted, ACSP Annual Conference, 2026.

Xuechun Yu. *From Installed to Inhabited: Measuring the Gap between Potential and Actualized Affordances in Children's Parks through User-Generated Content.* Abstract accepted, ACSP Annual Conference, 2026.

Research Experience

GISense Lab, The University of Texas at Austin — Austin, TX, USA
Research Collaborator, GeoAI Research — *Apr 2026 – Present*

- Lead a Detroit soundscape mapping project using street-view imagery to generate audio and estimate soundscape perceptions; prepare a first-author manuscript.
- Reproduce, fine-tune, and evaluate multimodal models, including SounDiT and image2sound, integrating acoustic, visual, and geospatial data.
- Evaluate generated audio against field recordings for acoustic similarity and consistency in downstream perceptual estimates.
- Contribute to drafting research project proposals in collaboration with the lab team.

China Academy of Urban Planning and Design (CAUPD), Shanghai Branch — Shanghai, China

Optimizing Urban Assessment Indicators for Child-Friendly Space Development — *Dec 2024 – May 2025*
Core Team Member · Funded by UNICEF

- Examined how child-friendly spatial criteria could be integrated into China's national urban assessment framework.
- Analyzed urban assessment indicators across multiple years to evaluate their relevance and sensitivity to child-friendly spatial outcomes.
- Proposed age-inclusive spatial indicators to inform recommendations for UNICEF-supported urban assessments.

National Best Practices in Child-Friendly Urban Space — *Oct 2024 – Jan 2025*
Core Team Member · Funded by Ministry of Housing and Urban-Rural Development (MOHURD)

- Managed case selection and drafting for three national compilations featuring 116 selected practices for child-friendly urban spaces.

- Built a database of local government submissions, developed screening criteria, coordinated expert review, and synthesized transferable lessons.
- Drafted content for compilations jointly issued by MOHURD and two other national bodies and circulated to provincial authorities nationwide for reference in local practice.

Evaluation of Child-Friendly City Development in Quzhou *Aug 2024 – Feb 2025*

Team Lead · Funded by Quzhou Women's Federation

- Led a three-person team evaluating Quzhou's child-friendly city initiative through surveys, field observation, indicator-based assessment, and comparative case analysis.
- Coordinated meetings with more than ten agencies and organizations, including the housing bureau and the Women's Federation; arranged field visits with community administrators.
- Conducted fieldwork in at least 30 communities, as well as schools and public facilities, to assess spatial conditions and user experiences.
- Produced a visual report on transferable practices and policy gaps in child-friendly, intergenerational public spaces.

Spatial Planning for Cross-Boundary Collaborative Zones in Shanghai Metropolitan Area *Apr 2024 – Mar 2025*

Core Team Member · Funded by Shanghai Municipal Administration of Planning and Natural Resources

- Developed an evaluation framework for integrated high-quality development in cross-jurisdictional areas, drawing on international comparative case studies.
- Led research on spatial optimization strategies for ecology, industry, and public service provision in the Jiading-Kunshan-Taicang border region.

Professional Experience

China Academy of Urban Planning and Design (CAUPD), Shanghai Branch Shanghai, China

Urban Planner *Apr 2023 – Jul 2025*

- Analyzed international metropolitan trends and coordination needs across 14 municipalities; contributed spatial and industrial strategies to the Shanghai Metropolitan Area Territorial Spatial Master Plan.
- Led the cultural strategy component of the Hangzhou Internationalization Pathway Study, proposing four action plans for cultural infrastructure.
- Helped develop plans for 60 sites across six types of public space, coordinating information across departments and assessing funding, site constraints, and managers' willingness to make changes.
- Conducted fieldwork in Shanghai and Wenzhou to examine children's access to outdoor spaces and community services.

Shenzhen University Institute of Urban Planning and Design Shenzhen, China

Assistant Planner *Apr 2018 – May 2020*

- Developed the initial concept and coordinated design revisions with international lead designers for the Longgang River Urban Design International Competition (team awarded 2nd place).
- Conducted urban positioning and industrial analysis for the Shenzhen East Station urban renewal project; reviewed local planning policies and design codes to inform proposals.

Skills & Languages

Programming & Data Analysis: Python, pandas, NumPy, XGBoost, SHAP

Geospatial Analysis: ArcGIS Pro, spatial data processing, spatial regression, MGWR, geovisualization

Urban Data Methods: Natural language processing, user-generated content analysis, audio preprocessing, acoustic feature extraction, multimodal data processing

Design & Visualization: Adobe Illustrator, Photoshop, InDesign, AutoCAD, Rhino, SketchUp

Languages: Mandarin (native); English (TOEFL iBT 5.0/6.0; IELTS 7.0/9.0)

Certification: Junior Engineer (certified by MOHURD)

Honors

Longgang River Urban Design International Competition, 2nd Place 2019

Innovation and Entrepreneurship Practice Award, Dalian University of Technology 2015